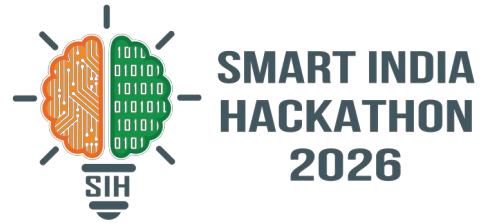


SMART INDIA HACKATHON 2026



NISHAN-PQ

Every decrypted copy leaves a verifiable mark.

PROBLEM STATEMENT ID	SIH26237
PROBLEM STATEMENT TITLE	Cryptographic Attribution and Immutable Decryption Provenance for Multi-Recipient Encrypted Document Distribution
THEME	Blockchain & Cybersecurity
PS CATEGORY	Software
TEAM ID	[ENTER TEAM ID]
TEAM NAME (REGISTERED ON PORTAL)	[ENTER REGISTERED TEAM NAME]

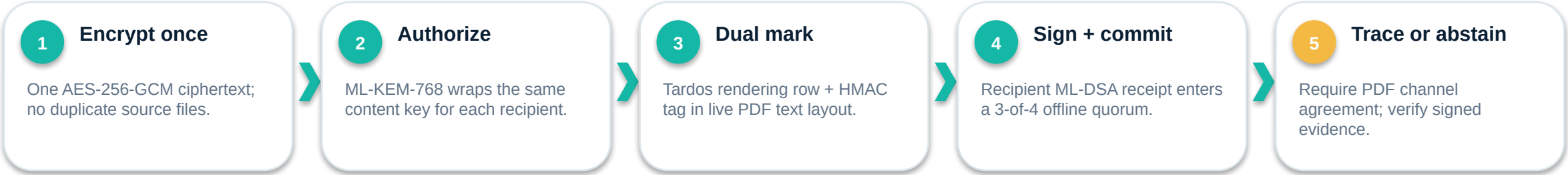


IDEA TITLE

[TEAM NAME]

NISHAN-PQ At-decryption leak attribution for air-gapped document distribution

PROPOSED SOLUTION | Bind two separately encoded PDF fingerprints + one recipient signature + one quorum receipt to every decrypted session before release.



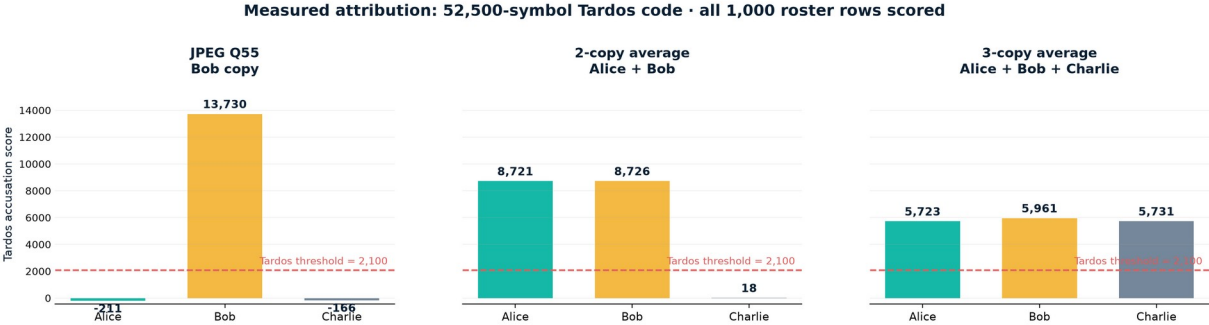
MEASURED DIFFERENTIATION

- 30/30 JPEG-Q55 sessions → exact signed release; 0 extra roster rows.
- Score $13,433 \pm 128$ vs $Z=2,100$; 30 codebooks $\times$ 4: all five 120/120, 0 innocent.
- Disjoint or partial visual/layout transplant → conflict → no accusation.

41.6+ dB

REAL PQC

FULLY OFFLINE



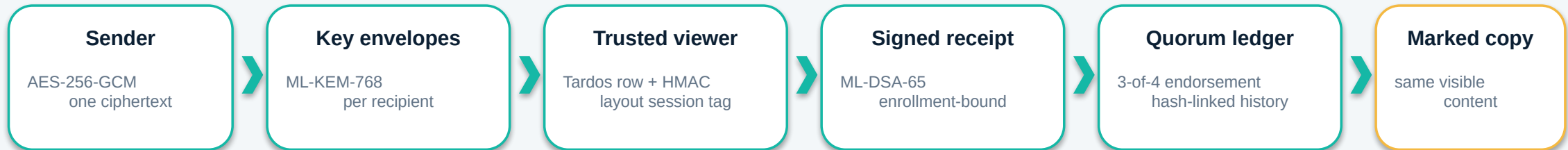
Measured prototype run · 10 Sep 2026 · all 1,000 rows scored · real physical captures pending

TECHNICAL APPROACH

[TEAM NAME]

One trust path; every security claim maps to a concrete mechanism

AIR-GAPPED TRUST BOUNDARY



LEAK → dual-channel consensus → trace / ABSTAIN → enrollment + signature + quorum verification

WORKING PROTOTYPE

- OpenSSL 3.6 · PyMuPDF · OpenCV ORB/RANSAC · full Tardos scorer
- Signed enrollment root rejects substitute keys; PDF conflicts fail closed

HARDENED DEPLOYMENT PATH

- Signed viewer + TPM/HSM keys + offline CA-bound identity
- Separate Fabric peers + external PQ head witness; no public chain or cryptocurrency

FEASIBILITY AND VIABILITY

[TEAM NAME]

IMPLEMENTED & MEASURED

- One encrypted broadcast package; three recipient key envelopes
- Tardos rendering row + domain-separated HMAC layout tag before release
- Searchable PDF + 52,500 symbols; all 1,000 rows scored against $Z = 2,100$
- Transplant + forged receipt stop; witness catches full-ledger rollback

41.87

MEAN PSNR · 5
PDFs

25 / 25

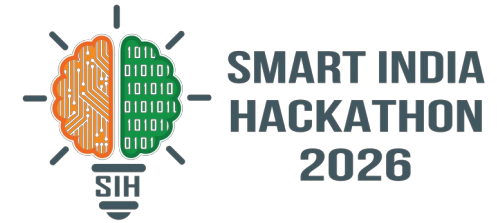
ATTACK RULES MATCH

120/120

CODE TRIALS ·
ALL 5

₹0 licences · commodity computer · no network service

Every number is generated by the reproducible demo; no hand-entered result.



POTENTIAL CHALLENGES → PREDECLARED VALIDATION GATES

HIGH

Print / photo

Synthetic perspective passes; hardware pending

GATE 150/300 dpi + phone-angle matrix

HIGH

PDF removal

Both carriers remain removable

GATE Require agreement; controlled viewer; bind to content

HIGH

Proof scope

JPEG has 11.2% marking violations

GATE Report condition per attack; add error-tolerant sync

MED

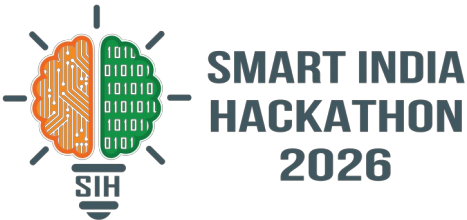
Trust custody

Authority controls marks; demo keys co-located

GATE Buyer-involved protocol; HSM keys; separate peers

Code theorem is conditional: it is never transferred silently to a carrier attack.

IMPACT AND BENEFITS



[TEAM NAME]

TARGET USER

Defence document controller & leak investigator

DECISION IMPROVED

“Which authorized decryption session produced this leak, and what verifiable evidence supports it?”

01 Operational accountability

Moves an investigation from an entire distribution list to one or more signed sessions.

02 Air-gap compatible

No cloud KMS, hosted API, public blockchain, token or internet dependency.

03 Post-quantum migration

Uses standardized ML-KEM and ML-DSA at the key-delivery and receipt layers.

04 Controlled adoption

Integrates as a viewer + SDK around existing document repositories and access policy.

ADOPTION AND SCALE PATH

PILOT

25 users · synthetic/unclassified corpus

SUCCESS: Trace rate, false accusation, latency

SECURE ENCLAVE

Separate validators · offline CA · HSM keys

SUCCESS: Print-scan and coalition attack matrix

ENTERPRISE

Existing DMS connectors · audited Fabric network

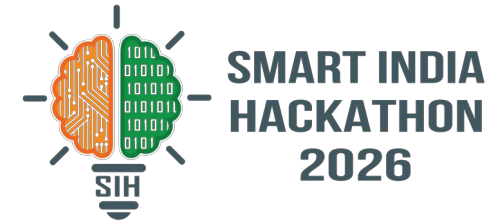
SUCCESS: Policy, monitoring and revocation at scale

RESEARCH AND REFERENCES

[TEAM NAME]

PRIMARY SOURCES & FOUNDATIONAL WORK

- 1 SIH26237 + SIH 2026 official guidelines**
sih.gov.in/sih2026PS · [/letters/SIH2026-Guidelines-College-SPOC.pdf](#)
- 2 NIST FIPS 203 / 204 — ML-KEM / ML-DSA**
csrc.nist.gov/pubs/fips/203/final · [/fips/204/final](#)
- 3 Wang et al. — T-Tracer (BSCI / ACM, 2022)**
doi.org/10.1145/3494106.3528674
- 4 Tardos — optimal probabilistic fingerprint codes**
renyi.hu/~tardos/fingerprint.pdf
- 5 Brassil et al. — text-layout document marking**
doi.org/10.1109/49.464718
- 6 Lu/Liao dual marks · Barreto et al. transplant attack**
doi.org/10.1109/83.951542 · doi.org/10.1109/ICIP.2001.958536
- 7 Working prototype + reproducible evidence**
[INSERT PUBLIC REPOSITORY LINK BEFORE UPLOAD]



PRIOR-ART BOUNDARY

KNOWN / PRESCRIBED

Tardos, layout marks, dual watermarking and transplant attacks already exist; SIH26237 prescribes PQ signatures + DLT.



OUR MEASURED CONTRIBUTION

Conflict-aware dual-domain evidence: Tardos + HMAC layout identities share one signed session; editable PDFs require agreement and transplants abstain.

Known gate: real print/photo matrix + content-bound carrier